

浙江大学



Lab Practice Report for Undergraduate

Name:

College:

ZheJiang University

Department:

Science

Major:

Marine Science

Student NO.:

Instructor:

Yanan Di

Date: 2024.11.30

Lab Practice Report of Zhejiang University

(此页可在 <http://bksy.zju.edu.cn/office/> 下载)

Course Name: Marine biology Type of Practice: dissect

Name of Practice: Anatomy of fish and the blood extraction

Student Name: _____ Major: Marine biology

Student Number: _____

Team Member: Zhiang Liu

Instructor: Yanan Di

Practice Location: 241 Practice Date: 2024.11.30

1. Aims and Requirement of the Practice (Compulsory)

(1) Learn the methods to humanely remove blood samples from anesthetized/alive fish.

Demonstrate gentle handling techniques of fish when alive fish are used.

(2) Learn about fish biology and complete a live or a virtual dissection of fish.

(3) Identify the external anatomy of fish; describe the function of important external features.

(4) Learn the fish blood preservation techniques.

(5) Practice contents extraction procedure from blood samples.

2. Background information of the Practice

Fish are vertebrates, including hagfish, lampreys, cartilaginous and bony fish. All of them have fins and most of them have scales. They are cold blood animals and lay eggs. Fish have similar structures with human.

In order to better live in the aquatic environment, fish developed identical circulatory system. Fishes have a one-way/single circulatory pathway through the heart. All the blood leaving the heart passes through at least two sets of capillaries; the capillaries of gills and the capillaries of body organ. The blood in the circulatory system transports oxygen to the tissues, and removes carbon dioxide from them. The dorsal aorta passes caudally beneath the vertebral column, giving

rise along the way to paired segmental intercostal, renal, and spinal arteries as well as unpaired arteries to the gas bladder and viscera. Posterior, where the paired ribs fuse to form the haemal arches, the dorsal aorta enters the haemal arch and passes into the tail. The caudal vein also lies within the haemal arches beneath the aorta. Branches of the dorsal aorta supply arterial blood to the head, body musculature, swim bladder, and all internal organs such as the gut, liver, gonads, and kidney.

3. Equipment and materials

- (1) Fish
- (2) Microcentrifuge tubes, glass beakers for washing purpose, washing bottles
- (3) Dissection set, sterile syringe and needles, timer
- (4) Marker pen pencils etc.

4. Procedures

- (1) Select one fish
- (2) Observing samples to identify the basic external morphology. Measure the size and make the record or take pictures.

- (3) Blood extraction

We choose Caudal vein method to get blood sample.

Insert the needle into the musculature perpendicular to the ventral surface of the fish until the spine is reached or blood enters the syringe. If contact with the spine is made withdraw the needle slightly. Collected blood in the microcentrifuge tubes.

- (4) Blood contents preservation

① Mix the blood with the SCS (1:1) mark the tube (6 tubes totally)

② Select 3 tubes gently homogenize and centrifuge at 4 °C 10000g 5min. preserve at -20°C

③ Select 3 tubes save at 4 °C.

5. Questions

- (1) Did you extract fish blood easily from the caudal vein? Why it can be easy or not?

It was not simple, although the classmate pressed the fish's body, the fishtail was still struggling, breaking three needles, and finally holding the fish's tail before the blood was

successfully drawn

- (2) What is the function of SCS you have used in this practice?

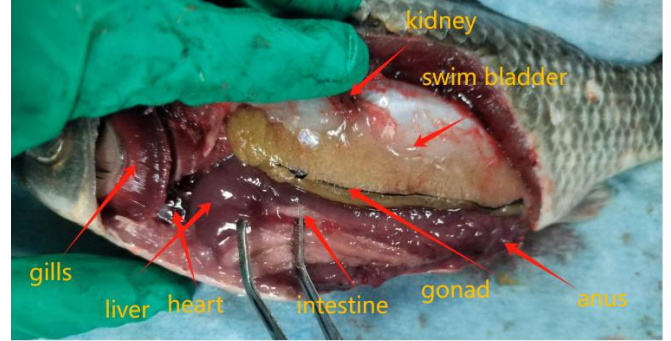
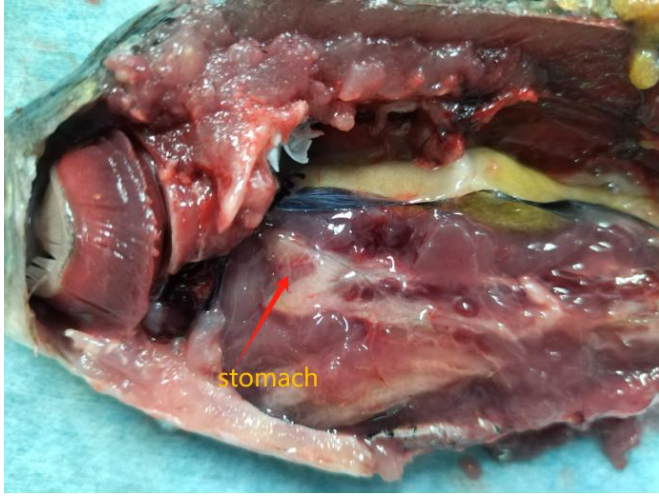
Anticoagulant to prevent fish blood from clotting

- (3) Take pictures of your samples; identify the internal and external structures and functions.

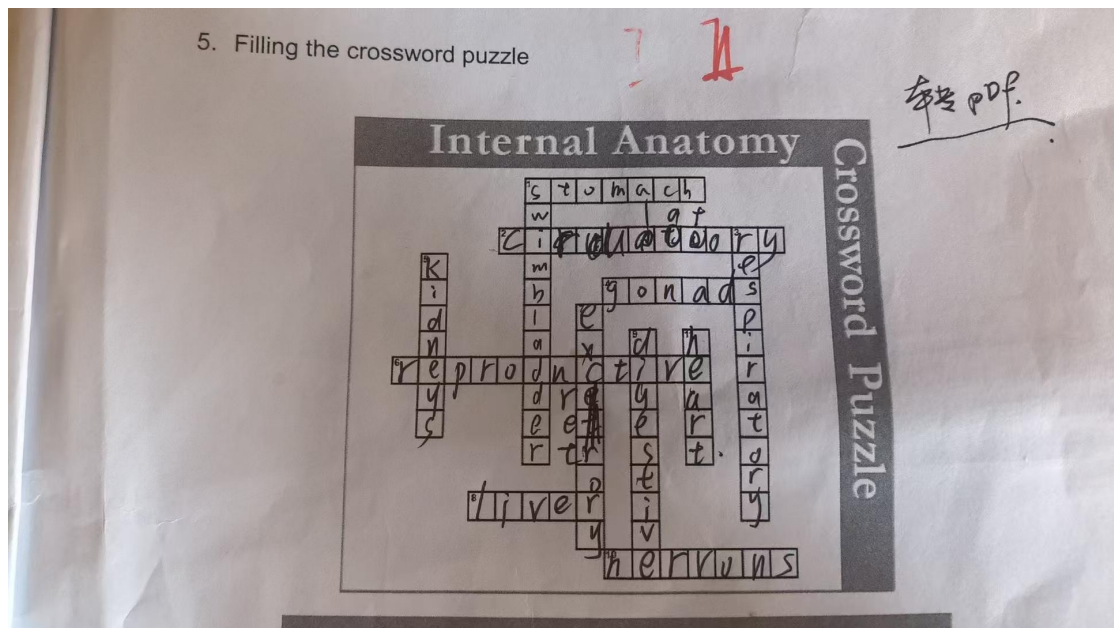
Size(18.21 cm x 8.11 cm x 3.95 cm)

Functions:

structures	functions
Stomach	Digestive sites
Heart	Blood supply
intestine	Absorb nutrients from food
Gills	Help breathe
Swim bladders	Help float in the water
digestive gland	Secrete digestive enzymes
Liver	Secrete various enzymes
kidney	Filter toxins



(4) Fill the crossword puzzle



注：不同类型的实验课对实验报告可有不同要求，各个学科的实验报告可以根据自己的学科特点做适当的调整，但上述基本内容中的第一、二、三、六条为必须填写的内容。

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装订线

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